

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM watsonx Orchestrate”

Project Tile – AdmissionGPT

Domain of Project – Education

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Problem Statement: AdmissionGPT

Choosing the right college and course is a challenging process for students and parents due to scattered information about colleges, admission criteria, cutoffs, fees, rankings, placements, and eligibility requirements. Students often find it difficult to compare colleges, analyze previous year admission trends, and make decisions according to their academic performance, interests, budget, and career goals.

The objective is to build an AI-powered Admission Agent that provides personalized college recommendations and admission guidance. The system retrieves reliable admission data such as rankings, accreditation, eligibility criteria, opening and closing ranks, category-wise cutoffs, state quota details, fees, hostel information, and placement records. It helps students shortlist suitable colleges, compare options, estimate admission chances, and receive guidance throughout the admission process.

The solution uses a multi-agent approach including an Admission Knowledge Agent for collecting college data, a College Recommendation Agent for suggesting suitable colleges and branches, an Admission Advisory Agent for counseling and application support, and a College Comparison Agent for decision-making. A visualization dashboard displays college comparisons, cutoff trends, admission probability, deadlines, and student progress to make the admission journey easier and more informed.

Technology Used

Tech Stack (Mandatory)

Use **any one** of the following approaches:

- LangFlow + IBM watsonx.ai + Granite Models
- IBM Cloud Agentic Lab with IBM Granite Models
- Replicate with LangChain + RAG + IBM Granite Models

I used IBM watsonx.ai + Granite Models

Proposed Solution -

Proposed Solution: **AdmissionGPT**

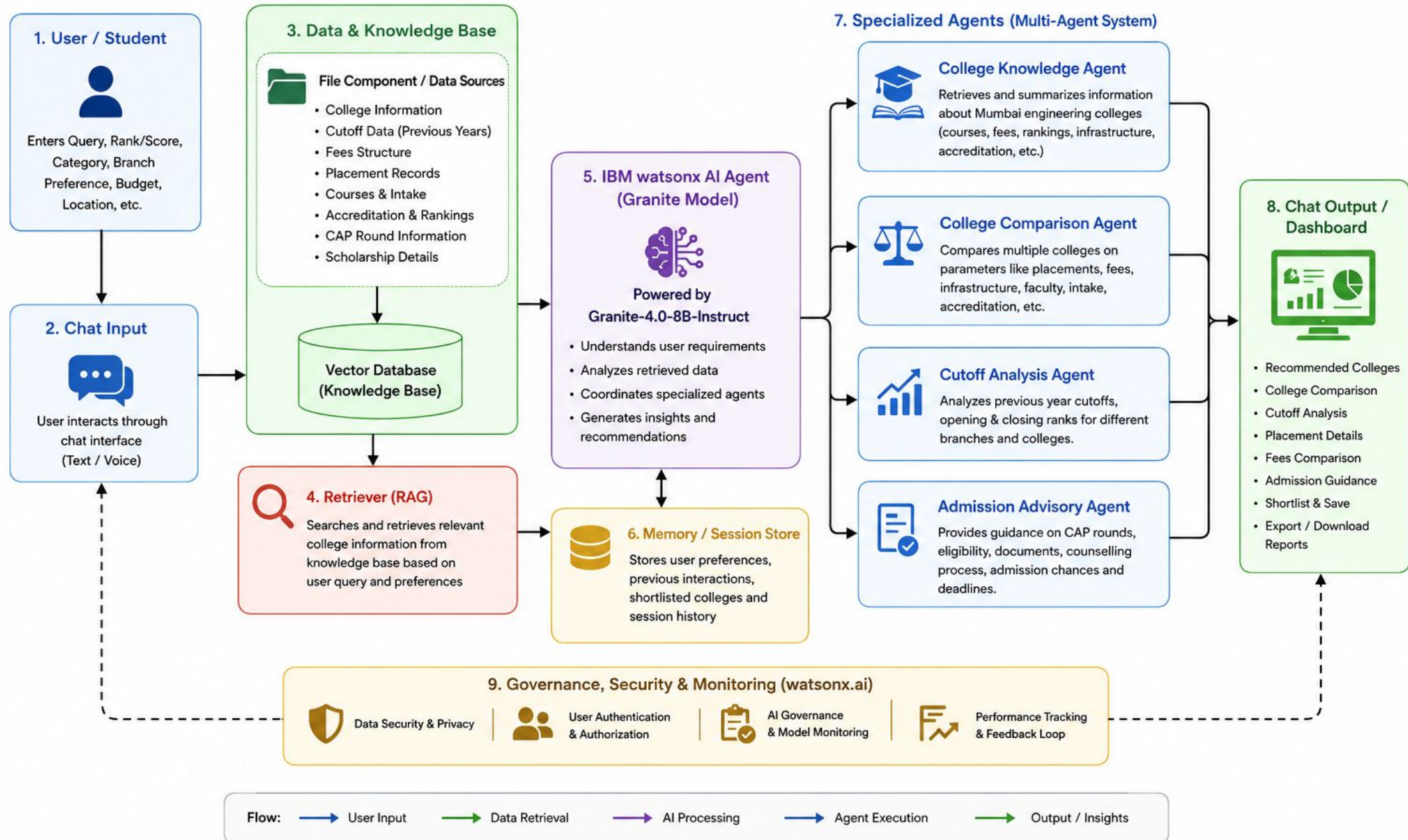
The proposed solution is an AI-powered Engineering College Comparison Agent that helps students compare Mumbai engineering colleges and make informed admission decisions. The system uses a RAG pipeline to collect reliable data from college websites and educational databases, providing information about rankings, fees, placements, accreditation, eligibility, and previous year cutoff trends.

The system includes AI agents such as a **College Knowledge Agent** for retrieving college details, a **College Recommendation Agent** for suggesting suitable colleges based on rank, category, branch, and preferences, and a **College Comparison Agent** for analyzing fees, placements, infrastructure, and academics. A dashboard displays college comparisons, cutoff analysis, placement details, and personalized recommendations using IBM [watsonx.ai](https://www.ibm.com/watsonx/ai) models.

IBM watsonx component Used -

1. **Chat Input** – Interface where users enter their queries, preferences, academic details, rank/score, category, and desired engineering branch.
2. **IBM watsonx AI Agent** – Processes user input using Granite models to generate personalized college recommendations and comparison insights.
3. **Name of IBM Granite model – Granite-4.0-8B-Instruct** → Fast and efficient for real-time college analysis, ranking comparison, and admission guidance.
4. **Chat output** – Displays AI-generated responses such as recommended colleges, cutoff analysis, placement details, fee comparison, and admission suggestions.
5. **File component** – Stores and retrieves college datasets, cutoff records, placement information, fees, and course details for comparison and analysis.
6. **Vector Database / Retriever Component** – Used for RAG-based retrieval of relevant college information from stored datasets and provides accurate responses based on user queries.

Mumbai Engineering College Comparison Agent – Workflow



College database: Knowledge

[Agent chat](#) / [Manage agents](#) / AdmissionGPT

[Test agent](#)

[Deploy](#)



AdmissionGPT



Model : GPT-OSS 120B — Ope...

Profile

Knowledge

Toolset

Behavior

Channels [Preview](#)

Describe your agent, including its purpose, tools, and agents, to help other agents and users know when to use it.

[What makes a good description?](#) ?

Welcome message

Customize the welcome message from the agent on the home screen.

Welcome message

30/100

Hello, welcome to AdmissionGPT

Maximum 100 characters allowed in welcome message edit field.

Quick start prompts

Pre-set messages sent to the agent to start the conversation.

[Add prompt](#) +

I scored 86 percentile in CET. Which Mumbai based engineering colleges can I get?



should I choose CSE, AI/ML, IT, or Electronics?



Recommend colleges near my home state



Agent style

Define how your agent interprets prompts and decide what action to take.

Build with AI

Talk to agent

[i](#) Running in debug mode

7	Fr. Conceição Institute of Technology (Pune campus, but Mumbai-city quota)	85-90 %	180 (B.Tech)	2.0 L – 2.6 L	5–6 L	–	civil Offe Mur quo goo stud pre Pun cam
8	Thadomal Shahani Engineering College (TSEC)	84-89 %	300 (B.Tech)	2.4 L – 2.9 L	5.5–6.5 L	150 (NIRF)	Larg vari acti pla
9	Mithibai College of Engineering (MCE)	85-90 %	150 (B.Tech)	2.7 L – 3.2 L	6–7 L	–	Par ren Mith stro com
10	Don Bosco Institute of Technology (DBIT) – Mumbai campus	84-88 %	120 (B.Tech)	2.3 L – 2.8 L	5–6 L	–	Emp holi dev goo in c eng

*Average salary figures are based on the most recent placement reports (2025-26) and represent the median for all branches combined. Individual branch outcomes (e.g., Computer Engineering vs. Civil) can differ markedly.

[Save as test](#) [i](#)

Type something...

CODE

The image shows a VS Code editor interface for a project named "bob". The Explorer sidebar on the left shows a file structure for "BOB" with folders like ".venv", "static", "css", "images", "js", and "templates", along with various HTML files and configuration files like ".env.example", ".gitignore", and "admission_guidance.db".

The main editor displays the file "app.py" with the following Python code:

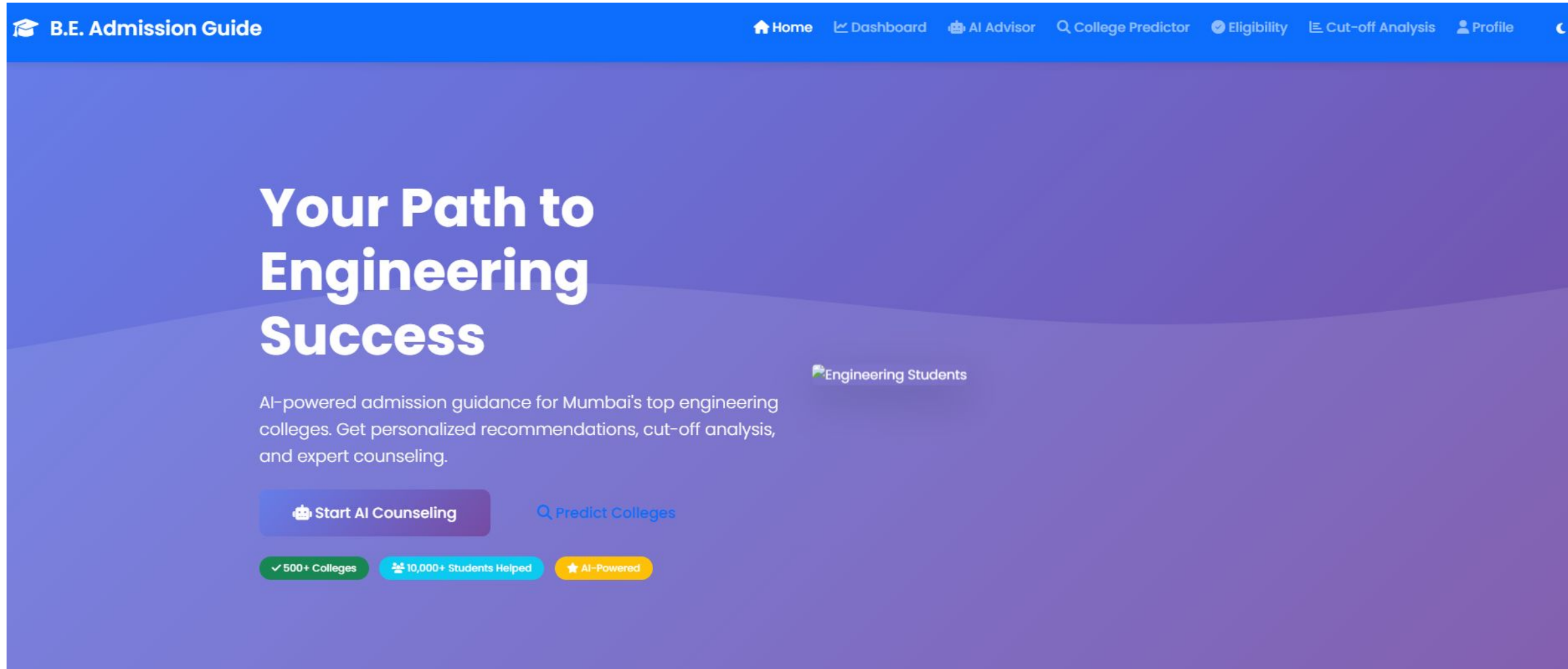
```
1 import os
2 import json
3 from flask import Flask, render_template, request, jsonify
4 from flask_cors import CORS
5 from dotenv import load_dotenv
6 from ibm_watsonx_ai.foundation_models import Model
7 from ibm_watsonx_ai.metanames import GenTextParamsMeta
8 from datetime import datetime
9 import sqlite3
10
11 # Load environment variables
12 load_dotenv()
13
14 app = Flask(__name__)
15 app.secret_key = os.getenv('SECRET_KEY', 'your-secret-key')
16 CORS(app)
17
18 # =====
19 # AGENT_INSTRUCTIONS - Customize AI Admission Advisor
20 # =====
21 AGENT_INSTRUCTIONS = """
22 You are an expert B.E. (Bachelor of Engineering) Admi
```

The bottom panel shows the "TERMINAL" with the following output:

```
127.0.0.1 - - [24/Jun/2026 13:27:18] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [24/Jun/2026 13:27:18] "GET /static/js/main.js HTTP/1.1" 200 -
127.0.0.1 - - [24/Jun/2026 13:27:18] "GET /static/css/style.css HTTP/1.1" 200 -
127.0.0.1 - - [24/Jun/2026 13:27:18] "GET /favicon.ico HTTP/1.1" 404 -
```

On the right side, there is a sidebar titled "IBM BOB" with a button "Upgrade your plan". Below it, a section titled "Create an AI-powe..." shows "All tasks completed!" and a list of deployment instructions (Heroku, Railway, Render, PythonAnywhere). A "Perfect For" section lists target users: Engineering aspirants in Mumbai/Maharashtra, Students preparing for MHT-CET/JEE Main, Parents seeking admission guidance, and Education counselors and consultants. The status is "COMPLETE AND READY TO USE", version is "1.0.0", technology is "Python Flask + IBM Watsonx.ai + Bootstrap 5", and the AI model is "IBM Granite 13B Chat v2". A note states: "All features implemented, tested, and documented. The application is production-ready and can be deployed." At the bottom right, there is a "Start New Task" button.

INPUT SCREEN -I



INPUT SCREEN -II

Our Features

Everything you need for B.E. admission guidance



AI Admission Advisor

Chat with our AI counselor powered by IBM Watsonx.ai for personalized guidance

[Try Now](#)



College Predictor

Predict your chances of admission based on marks, category, and preferences

[Predict](#)



Cut-off Analysis

Analyze historical cut-off trends and predict current year cut-offs

[Analyze](#)



Eligibility Checker

Check if you meet the eligibility criteria for engineering admission

[Check](#)



Admission Dashboard

Track your admission journey with interactive analytics and insights

[View](#)



Student Profile

Create and manage your profile for personalized recommendations

[Create](#)

Chat Input

 **AI Admission Advisor**
Powered by IBM Watsonx.ai Granite Model

Clear



Chat cleared! How can I help you today?

Quick Questions:

[Top Colleges](#) [CAP Process](#) [Documents](#) [Fee Structure](#)

i got 86 percentile in CET, want admission in Computer /open , suggest me college

✓

Send

Output

i got 86 percentile, suggest college

12k / 200k 0.06

i got 86 percentile in CET, want admission in Computer /open , suggest me college

Code



Perfect! With 86 percentile in MHT-CET, OPEN category, Computer Engineering, here are my detailed college recommendations:

 Task Completed



College Recommendations for 86 Percentile (MHT-CET, OPEN, Computer Engineering)

🎯 REALISTIC TARGET COLLEGES (Good Chances)

- 1. Terna Engineering College, Navi Mumbai**
 - Cut-off: ~85-90 percentile (OPEN)
 - Fees: ₹1.2-1.5 lakhs/year
 - Strong placement record, good infrastructure
 - Probability: HIGH
- 2. Atharva College of Engineering, Malad**
 - Cut-off: ~82-88 percentile (OPEN)
 - Fees: ₹1.3 lakhs/year
 - Decent placements, growing reputation
 - Probability: HIGH
- 3. Fr. Conceicao Rodrigues College of Engineering, Bandra**
 - Cut-off: ~84-89 percentile (OPEN)

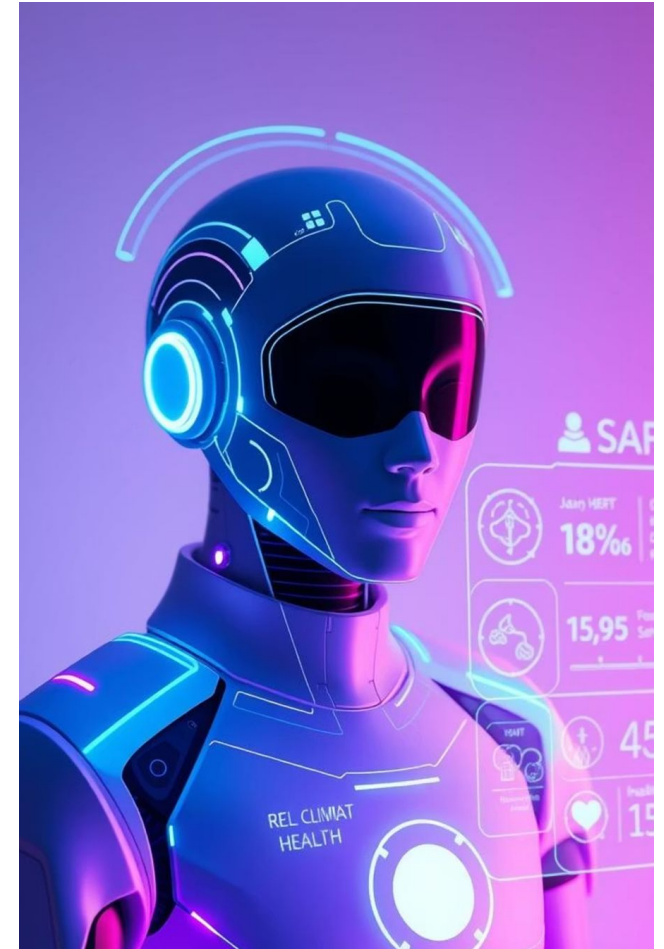
Role of Agentic AI in the solution

Agentic AI enables the Mumbai Engineering College Comparison Agent to work as an intelligent decision-support system rather than a basic search tool. Using IBM watsonx.ai and IBM Granite models, it coordinates multiple specialized agents to collect college data, analyze student preferences, and provide personalized college recommendations.

Each agent performs a specific role—such as retrieving college information, comparing colleges, analyzing cutoffs, and suggesting suitable options—while working together to generate accurate results. This multi-agent approach helps students efficiently evaluate colleges based on rank, category, branch preference, fees, placements, and location.

Agentic AI provides context awareness by understanding student requirements and adapting recommendations according to their academic profile and career goals. It enables real-time analysis of college data and improves decision-making through continuous interaction between agents.

Overall, Agentic AI transforms the system into a smart college admission assistant that provides personalized comparisons, reliable insights, and simplified guidance for students selecting engineering colleges.



Git Hub Link

<https://github.com/farhanasiddiqui-cmd/AdmissionGPT.git>

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Future Scope

Scope of the Agent –

The Mumbai Engineering College Comparison Agent is currently focused only on engineering colleges in Mumbai, providing students with detailed comparisons and recommendations based on available admission, academic, and placement data. In the future, the system can be expanded to cover engineering colleges across India.

1. Integration with Admission Portals & Real-Time College Data –

The system can be integrated with official admission portals and college databases to collect real-time information such as cutoff updates, seat availability, fees, course details, and admission notifications. This will enable the College Comparison Agent to provide accurate and updated recommendations based on current admission trends.

2. AI-Powered Voice Assistant & Multilingual Support –

Enhancing the system with a voice-enabled assistant and multilingual support will improve accessibility for students. Users can interact naturally in their preferred language, making the platform easier to use and helping students from different backgrounds make better college selection decisions.

Thank You!

Thank you for your time and interest.